



APPLIED MATHS

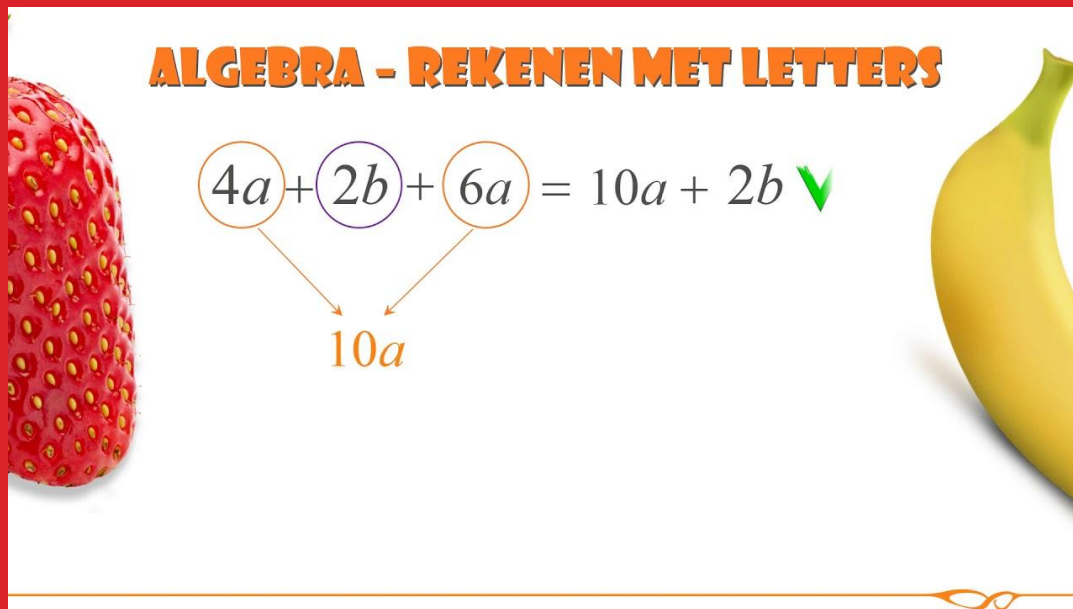
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Bachelor in de elektronica-ICT Brugge – 2025-2026



HOOFDSTUK 2 : REKENEN MET LETTERVORMEN

ALGEBRA - REKENEN MET LETTERS


$$\begin{array}{c} 4a + 2b + 6a = 10a + 2b \checkmark \\ \swarrow \quad \searrow \\ 10a \end{array}$$

Hoofdstuk 2: rekenen met lettervormen

Merkwaardige producten

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a - b)^2 = a^2 - 2ab + b^2$$

$$(a - b) \cdot (a + b) = a^2 - b^2$$

Binomium van Newton/Driehoek van Pascal

$$(a + b)^3 = 1 \cdot a^3 + 3 \cdot a^2 \cdot b + 3 \cdot a \cdot b^2 + 1 \cdot b^3$$

$$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$$

"The Binomial Theorem Formula"

Reduce the 'a' term and increase the 'b' term each time

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

The Sigma sign tells us to add up the terms

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

$$(a + b)^n = \binom{n}{0} a^n b^0 + \binom{n}{1} a^{n-1} b^1 + \binom{n}{2} a^{n-2} b^2 + \dots + \binom{n}{n} a^0 b^n$$

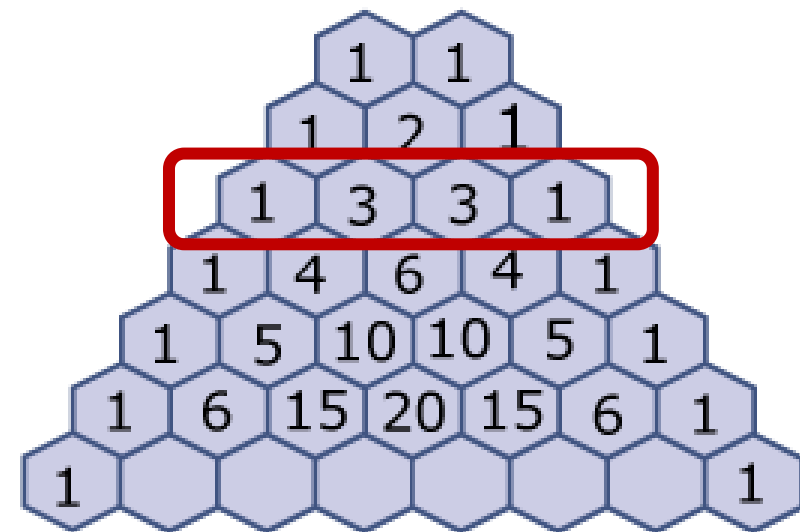
The Binomial Theorem

$$(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_k x^k + \dots + {}^nC_n x^n$$

$$= \sum_{k=0}^n {}^nC_k x^k$$

where ${}^nC_k = \frac{n!}{k!(n-k)!}$ and n is a positive integer

NOTE: there are $(n+1)$ terms



Hoofdstuk 2: rekenen met lettervormen

Euclidische deling of staartdeling

$$\frac{x^3 - x + 7}{x - 1} = \dots$$

$x^3 + 0x^2 - x + 7$	$x - 1$
$\begin{array}{r} x^3 - x^2 \\ \hline x^2 - x \\ x^2 - x \\ \hline 0 + 7 \end{array}$	$\begin{array}{r} x^2 + x \end{array}$

Hoofdstuk 2: rekenen met lettervormen

Euclidische deling of staartdeling

$$\frac{x^3 - x + 7}{x - 1} = \dots$$

$T(x)$	$N(x)$
$x^3 + 0x^2 - x + 7$	$x - 1$
$x^3 - x^2$	$x^2 + x$
$- \quad \quad \quad$	$Q(x)$
$x^2 - x$	
$x^2 - x$	
$- \quad \quad \quad$	
$0 + 7$	
$R(x)$	

$$\text{Dus } \frac{x^3 - x + 7}{x - 1} = x^2 + x + \frac{7}{x - 1}$$

Stop: als graad $R(x) < \text{graad } N(x)$

$$\text{Resultaat: } \frac{T(x)}{N(x)} = Q(x) + \frac{R(x)}{N(x)}$$

Hoofdstuk 2: rekenen met lettervormen

Regel van Horner

$$\frac{x^3 - x + 7}{x - 1} = \dots$$

Begin

	1	0	-1		7
1	↓				

Resultaat

	1	0	-1		7
1	↓	1	1		0
	1	1	0		7

Hoofdstuk 2: rekenen met lettervormen

Regel van Horner

$$\frac{x^3 - x + 7}{x - 1} = \dots$$

				$T(x)$
$N(x)$	1	0	-1	7
	↓	1	1	0
	1	1	0	7
$Q(x)$				$R(x)$

Interpretatie

$$\frac{T(x)}{N(x)} = Q(x) + \frac{R(x)}{N(x)} \text{ dus } \frac{x^3 - x + 7}{x - 1} = x^2 + x + \frac{7}{x - 1}$$

Opmerking: enkel bruikbaar als graad $N(x) = 1$

Hoofdstuk 2: rekenen met lettervormen

Discriminant

$$ax^2 + bx + c = a(x - x_1)(x - x_2)$$

$$\text{met } D = b^2 - 4ac \quad \text{en} \quad x_{1,2} = \frac{-b \pm \sqrt{D}}{2a}$$

Voorbeeld

$$2x^2 - 10x + 12 = 2(x - 2)(x - 3)$$

$$\text{Want } D = (-10)^2 - 4 \cdot 2 \cdot 12 = 4 \quad \text{dus} \quad x_{1,2} = \frac{10 \pm 2}{4} = 3 \text{ of } 2$$

Hoofdstuk 2: rekenen met lettervormen

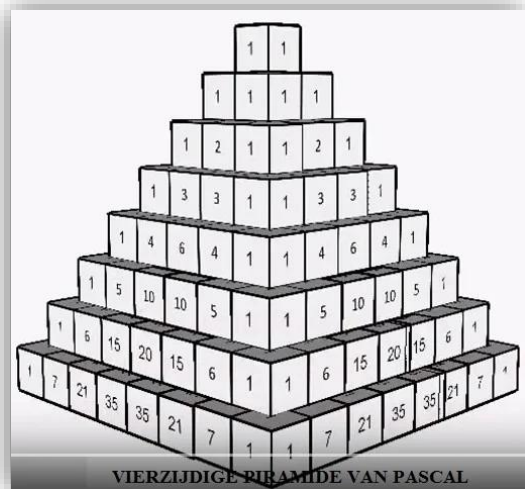
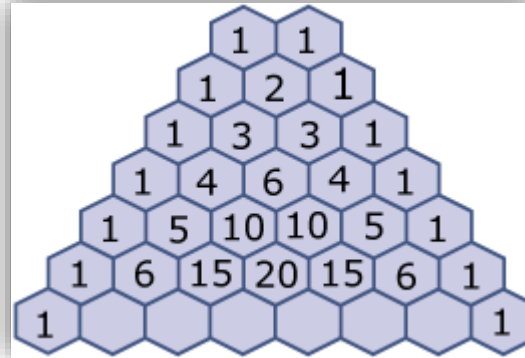
Merkwaardige producten

$$(a + b)^2 = a^2 + 2.ab + b^2$$

$$(a - b)^2 = a^2 - 2.ab + b^2$$

$$(a + b).(a - b) = a^2 - b^2$$

$$(x + a).(x + b) = x^2 + (a + b).x + b^2$$



$$(a + b)^3 = a^3 + 3.a^2b + 3.ab^2 + b^3$$

$$(a - b)^3 = a^3 - 3.a^2b + 3.ab^2 - b^3$$

$$(a \pm b)^3 = a^3 \pm 3.a^2b + 3.ab^2 \pm b^3$$

$$a^3 + b^3 = (a + b).(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a - b).(a^2 + ab + b^2)$$

$$(a \pm b)^4 = a^4 \pm 4.a^3b + 6.a^2b^2 \pm 4.ab^3 + b^4$$

Hoofdstuk 2: rekenen met lettervormen

Kwadratische veeltermen ontbinden

THE QUADRATIC FORMULA © CHILIMATH.COM

If $ax^2 + bx + c = 0$ but $a \neq 0$

then

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

DISCRIMINANT

- ➡ $b^2 - 4ac > 0$ two real solutions
- ➡ $b^2 - 4ac = 0$ one real solutions
- ➡ $b^2 - 4ac < 0$ zero real solutions

If the discriminant > 0 ,

$$\text{root1} = \frac{-b + \sqrt{(b - 4ac)}}{2a}$$

$$\text{root2} = \frac{-b - \sqrt{(b - 4ac)}}{2a}$$

If the discriminant $= 0$,

$$\text{root1} = \text{root2} = \frac{-b}{2a}$$

If the discriminant < 0 ,

$$\text{root1} = \frac{-b}{2a} + i \frac{\sqrt{-(b - 4ac)}}{2a}$$

$$\text{root2} = \frac{-b}{2a} - i \frac{\sqrt{-(b - 4ac)}}{2a}$$



EINDE HOOFDSTUK 2 : REKENEN MET LETTERVORMEN

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